

BNPDFormer Best Checkpoint Metrics

dense_i1_a1_prefix8 (unsupervised, epoch 8)
Official retained recipe: FactoryBN_dense_f1_p80.json

Protocol. Dataset dense_i1 ($N=38$, $F=27$); episode split 140/28/36 (seed 42); windows 20826/4331/5354. Input $T_{\text{in}}=30$ min, horizon $H=15$ min. A report is correct iff the station matches *and* $|\hat{s} - s| \leq 3$ min. Thresholds are selected on validation with report precision ≥ 0.80 , then frozen on test. Occupancy-grid F1 and IoU-event F1 are not checkpoint metrics and are omitted (not recorded in the retained notes).

Table 1: Station-level event reports (primary task A.1). Checkpoint metric is report F1.

Group	Metric	Validation	Test
Report (primary)	Precision P	0.883	0.852
	Recall R	0.881	0.856
	F1	0.882	0.854
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.633	0.455
	Ongoing $R_{\text{on}} (s=0)$	0.920	0.911
Support	n_{true} upcoming	98	112
	n_{true} ongoing	624	828
	n_{true} total	722	940

Table 2: Order remaining time and process-cause heads (do not override report F1).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	≈ 12.0	10.5
Process cause (6 classes)	Overall accuracy	—	0.965
	Macro recall (support>0)	0.774	0.971
	Majority-class baseline (train)	0.41	
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.956
	<i>material_shortage</i>	—	0.988
	<i>starved_upstream</i>	—	0.949
	<i>queue_buildup</i>	—	0.992
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Taken from retained notes for prefix8; last_metrics.json is not in this workspace, so who_*, start_mae, dur_mae, occupancy-grid F1 and IoU-event F1 are not invented. (2) Cause labels are process-heuristic: report test macro 0.971 together with val macro 0.774 and the train majority baseline 0.41; this is not a causal claim. (3) Upcoming recall 0.455 (112 test events) remains the open early-warning failure mode. (4) Rejected on the same protocol: precursor head (F1 0.827), score regression (0.850), STGNPP auxiliary (0.849).

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dense_i1_12_3_start5_min8_cause4_opt_main_seed42
Recipe: FactoryBN_dense_12_3_start5_min8_opt.json; best epoch 39

Protocol. Same dense_i1 pack and episode split as above. Event horizon capped at start ≤ 5 min; Min8 occupancy filter. Checkpoint metric is will15 F1 (report / who track it within ~ 0.001). Thresholds selected on validation with report precision ≥ 0.80 , then frozen on test. Numbers from last_metrics.json.

Table 3: Station-level event reports (start ≤ 5). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.837	0.794
	Recall R	0.891	0.817
	F1	0.863	0.805
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.739	0.539
	Ongoing $R_{\text{on}} (s=0)$	0.969	0.937
Support	n_{true} upcoming	—	336
	n_{true} ongoing	—	766
	n_{true} total	—	1102

Table 4: Order remaining time and process-cause heads (start ≤ 5).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	12.3	10.9
Process cause (4 active / 6 classes)	Overall accuracy	0.956	0.965
	Macro recall (support >0)	0.963	0.967
	Majority-class baseline (train)		0.409
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.941
	<i>material_shortage</i>	—	0.976
	<i>starved_upstream</i>	—	0.966
	<i>queue_buildup</i>	—	0.984
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.805; report F1 0.804). (2) Val support counts are not stored in last_metrics.json. (3) Upcoming recall 0.539 on 336 test events is the main early-warning gap at this horizon.

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dense_i1_12_3_start10_min8_cause4_opt_main_seed42
Recipe: FactoryBN_dense_12_3_start10_min8_opt.json; best epoch 41

Protocol. Same pack / split as above. Event horizon capped at $\text{start} \leq 10$ min; Min8 occupancy filter. Checkpoint metric is will15 F1. Thresholds selected on validation with report precision ≥ 0.80 , then frozen on test. Numbers from `last_metrics.json`.

Table 5: Station-level event reports ($\text{start} \leq 10$). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.831	0.806
	Recall R	0.840	0.761
	F1	0.835	0.783
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.711	0.528
	Ongoing $R_{\text{on}} (s=0)$	0.960	0.936
Support	n_{true} upcoming	—	604
	n_{true} ongoing	—	768
	n_{true} total	—	1372

Table 6: Order remaining time and process-cause heads ($\text{start} \leq 10$).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	12.6	11.2
Process cause (4 active / 6 classes)	Overall accuracy	0.958	0.962
	Macro recall (support>0)	0.968	0.965
	Majority-class baseline (train)		0.409
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.926
	<i>material_shortage</i>	—	0.988
	<i>starved_upstream</i>	—	0.962
	<i>queue_buildup</i>	—	0.984
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are `report_*` at the best will15 epoch; test primary is `will15_*` (F1 0.783; report F1 0.778). (2) Upcoming pool grows vs `start5` ($n_{\text{up}}=604$), but R_{up} stays near 0.53. (3) Of upcoming test events: $\text{start} \leq 5$: 336; $6 \leq \text{start} \leq 10$: 268.

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dense_i1_12_3_start15_min8_cause4_opt_main_seed42
Recipe: FactoryBN_dense_12_3_start15_min8_opt.json; best epoch 42

Protocol. Same pack / split as above. Event horizon capped at $\text{start} \leq 15$ min; Min8 occupancy filter. Checkpoint metric is will15 F1. Thresholds selected on validation with report precision ≥ 0.80 , then frozen on test. Numbers from `last_metrics.json`.

Table 7: Station-level event reports ($\text{start} \leq 15$). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.841	0.806
	Recall R	0.835	0.742
	F1	0.838	0.772
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.722	0.534
	Ongoing $R_{\text{on}} (s=0)$	0.960	0.925
Support	n_{true} upcoming	—	708
	n_{true} ongoing	—	770
	n_{true} total	—	1478

Table 8: Order remaining time and process-cause heads ($\text{start} \leq 15$).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	11.0	9.7
Process cause (4 active / 6 classes)	Overall accuracy	0.967	0.974
	Macro recall (support>0)	0.978	0.978
	Majority-class baseline (train)		0.409
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.978
	<i>material_shortage</i>	—	0.988
	<i>starved_upstream</i>	—	0.962
	<i>queue_buildup</i>	—	0.984
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are `report_*` at the best will15 epoch; test primary is `will15_*` (F1 0.772; report F1 0.768). (2) Hardest setting: farthest starts included ($n_{\text{up}}=708$; $\text{start}>10$: 104), test will15 F1 drops to 0.772. (3) Upcoming bucket recalls (test): $\text{start} \leq 5$: 0.574; $6 \leq \text{start} \leq 10$: 0.522; $\text{start}>10$: 0.433.

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ablation_nograph_start5_seed42

Recipe: FactoryBN_dense_struct_nograph_start5_min8.json; best epoch 19

Protocol. Same dense_i1 pack and episode split as the full baseline. Ablation: graph_mode=identity (no GNN edges). Event horizon capped at start ≤ 5 min; Min8 occupancy filter. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 9: Station-level event reports (start ≤ 5 , nograph). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.823	0.832
	Recall R	0.658	0.678
	F1	0.731	0.747
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.115	0.101
	Ongoing $R_{\text{on}} (s=0)$	0.937	0.930
Support	n_{true} upcoming	—	336
	n_{true} ongoing	—	766
	n_{true} total	—	1102

Table 10: Order remaining time and process-cause heads (start ≤ 5 , nograph).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	18.2	17.0
Process cause (4 active / 6 classes)	Overall accuracy	0.866	0.876
	Macro recall (support>0)	0.886	0.889
	Majority-class baseline (train)	0.409	
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.971
	<i>material_shortage</i>	—	1.000
	<i>starved_upstream</i>	—	0.852
	<i>queue_buildup</i>	—	0.734
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.747; report F1 0.746). (2) From-scratch ablation (50 epochs, no init_ckpt); best epoch 19. (3) Upcoming recall 0.101 on 336 test events (vs full baseline at the same start horizon).

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ablation_nograph_start10_seed42

Recipe: FactoryBN_dense_struct_nograph_start10_min8.json; best epoch 9

Protocol. Same dense_i1 pack and episode split as the full baseline. Ablation: graph_mode=identity (no GNN edges). Event horizon capped at start ≤ 10 min; Min8 occupancy filter. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 11: Station-level event reports (start ≤ 10 , nograph). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.807	0.821
	Recall R	0.486	0.539
	F1	0.606	0.651
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.049	0.065
	Ongoing $R_{\text{on}} (s=0)$	0.894	0.909
Support	n_{true} upcoming	—	604
	n_{true} ongoing	—	768
	n_{true} total	—	1372

Table 12: Order remaining time and process-cause heads (start ≤ 10 , nograph).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	24.1	23.3
Process cause (4 active / 6 classes)	Overall accuracy	0.741	0.734
	Macro recall (support>0)	0.755	0.746
	Majority-class baseline (train)	0.409	
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.993
	<i>material_shortage</i>	—	0.963
	<i>starved_upstream</i>	—	0.738
	<i>queue_buildup</i>	—	0.290
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.651; report F1 0.649). (2) From-scratch ablation (50 epochs, no init_ckpt); best epoch 9. (3) Upcoming recall 0.065 on 604 test events (vs full baseline at the same start horizon).

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ablation_nograph_start15_seed42

Recipe: FactoryBN_dense_struct_nograph_start15_min8.json; best epoch 11

Protocol. Same dense_i1 pack and episode split as the full baseline. Ablation: graph_mode=identity (no GNN edges). Event horizon capped at start ≤ 15 min; Min8 occupancy filter. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 13: Station-level event reports (start ≤ 15 , nograph). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.819	0.851
	Recall R	0.435	0.482
	F1	0.568	0.616
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.028	0.037
	Ongoing $R_{\text{on}} (s=0)$	0.883	0.887
Support	n_{true} upcoming	—	708
	n_{true} ongoing	—	770
	n_{true} total	—	1478

Table 14: Order remaining time and process-cause heads (start ≤ 15 , nograph).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	21.4	20.8
Process cause (4 active / 6 classes)	Overall accuracy	0.796	0.794
	Macro recall (support>0)	0.790	0.787
	Majority-class baseline (train)	0.409	
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.993
	<i>material_shortage</i>	—	0.988
	<i>starved_upstream</i>	—	0.878
	<i>queue_buildup</i>	—	0.290
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.616; report F1 0.612). (2) From-scratch ablation (50 epochs, no init_ckpt); best epoch 11. (3) Upcoming recall 0.037 on 708 test events (vs full baseline at the same start horizon).

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ablation_nogroup_start5_seed42

Recipe: FactoryBN_dense_struct_nogroup_start5_min8.json; best epoch 28

Protocol. Same dense_i1 pack and episode split as the full baseline. Ablation: no station-group / semantic grouping features. Event horizon capped at start ≤ 5 min; Min8 occupancy filter. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 15: Station-level event reports (start ≤ 5 , nogroup). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.782	0.766
	Recall R	0.765	0.756
	F1	0.774	0.761
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.383	0.307
	Ongoing $R_{\text{on}} (s=0)$	0.962	0.941
Support	n_{true} upcoming	—	336
	n_{true} ongoing	—	766
	n_{true} total	—	1102

Table 16: Order remaining time and process-cause heads (start ≤ 5 , nogroup).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	13.2	12.7
Process cause (4 active / 6 classes)	Overall accuracy	0.932	0.943
	Macro recall (support>0)	0.952	0.955
	Majority-class baseline (train)	0.409	
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.956
	<i>material_shortage</i>	—	1.000
	<i>starved_upstream</i>	—	0.911
	<i>queue_buildup</i>	—	0.952
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.761; report F1 0.753). (2) From-scratch ablation (50 epochs, no init_ckpt); best epoch 28. (3) Upcoming recall 0.307 on 336 test events (vs full baseline at the same start horizon).

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ablation_nogroup_start10_seed42

Recipe: FactoryBN_dense_struct_nogroup_start10_min8.json; best epoch 44

Protocol. Same dense_i1 pack and episode split as the full baseline. Ablation: no station-group / semantic grouping features. Event horizon capped at start ≤ 10 min; Min8 occupancy filter. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 17: Station-level event reports (start ≤ 10 , nogroup). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.663	0.605
	Recall R	0.790	0.745
	F1	0.721	0.668
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.636	0.445
	Ongoing $R_{\text{on}} (s=0)$	0.934	0.947
Support	n_{true} upcoming	—	604
	n_{true} ongoing	—	768
	n_{true} total	—	1372

Table 18: Order remaining time and process-cause heads (start ≤ 10 , nogroup).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	12.5	11.6
Process cause (4 active / 6 classes)	Overall accuracy	0.928	0.938
	Macro recall (support>0)	0.944	0.953
	Majority-class baseline (train)		0.409
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.956
	<i>material_shortage</i>	—	1.000
	<i>starved_upstream</i>	—	0.890
	<i>queue_buildup</i>	—	0.968
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.668; report F1 0.651). (2) From-scratch ablation (50 epochs, no init_ckpt); best epoch 44. (3) Upcoming recall 0.445 on 604 test events (vs full baseline at the same start horizon).

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ablation_nogroup_start15_seed42

Recipe: FactoryBN_dense_struct_nogroup_start15_min8.json; best epoch 50

Protocol. Same dense_i1 pack and episode split as the full baseline. Ablation: use_grouped_embed=false (no station-group embedding). Event horizon capped at start ≤ 15 min; Min8 occupancy filter. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 19: Station-level event reports (start ≤ 15 , nogroup). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.620	0.567
	Recall R	0.760	0.690
	F1	0.683	0.623
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.609	0.395
	Ongoing $R_{\text{on}} (s=0)$	0.927	0.922
Support	n_{true} upcoming	—	708
	n_{true} ongoing	—	770
	n_{true} total	—	1478

Table 20: Order remaining time and process-cause heads (start ≤ 15 , nogroup).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	11.1	10.6
Process cause (4 active / 6 classes)	Overall accuracy	0.941	0.938
	Macro recall (support>0)	0.950	0.949
	Majority-class baseline (train)	0.409	
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.926
	<i>material_shortage</i>	—	1.000
	<i>starved_upstream</i>	—	0.911
	<i>queue_buildup</i>	—	0.960
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.623; report F1 0.604). (2) From-scratch ablation (50 epochs, no init_ckpt); best epoch is the last epoch (50), so this run did not plateau inside the budget. (3) Validation report precision at that epoch is 0.620, below the official $P \geq 0.80$ gate used by the full baseline. Upcoming recall 0.395 on 708 test events. Bucket recalls: start ≤ 5 : 0.446; $6 \leq \text{start} \leq 10$: 0.410; start>10: 0.192.

Structure ablation vs full baseline

Test will15, same dense_i1 split, seed 42

Reading. Full rows are the retained 12_3 runs (up to 100 epochs, best around epoch 39–42). Ablation rows are from-scratch, 50 epochs, empty `init_ckpt`. $\Delta F1$ is ablation minus full at the same start cap. R_{up} is report upcoming recall, not the checkpoint metric.

Table 21: Test will15 and auxiliary heads. Bold F1 is the checkpoint metric.

Arm	Start	P	R	F1	R_{up}	R_{on}	remain MAE
full	≤ 5	0.794	0.817	0.805	0.539	0.937	10.9
nograph	≤ 5	0.832	0.678	0.747	0.101	0.930	17.0
nogroup	≤ 5	0.766	0.756	0.761	0.307	0.941	12.7
full	≤ 10	0.806	0.761	0.783	0.528	0.936	11.2
nograph	≤ 10	0.821	0.539	0.651	0.065	0.909	23.3
nogroup	≤ 10	0.605	0.745	0.668	0.445	0.947	11.6
full	≤ 15	0.806	0.742	0.772	0.534	0.925	9.7
nograph	≤ 15	0.851	0.482	0.616	0.037	0.887	20.8
nogroup	≤ 15	0.567	0.690	0.623	0.395	0.922	10.6

Notes. (1) Only `nograph` keeps validation report precision ≥ 0.80 at the selected epoch (0.823 / 0.807 / 0.819). `nogroup` selected epochs sit at 0.782 / 0.663 / 0.620, so those checkpoints would be rejected by the full-model precision gate. (2) `nosem` (`ablation_nosem_start5_seed42`) aborted before epoch 1: a convolution weight of shape `[0, 80, 1, 1]`. `nopattern` was never started. (3) Cause macro recall stays near the full model for `nogroup` (0.955 / 0.953 / 0.949) and drops for `nograph`, especially `queue_buildup` (0.734 / 0.290 / 0.290 vs full ≈ 0.98).

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dense_i1_entity_noinfo_start5_min8_cold_ep50_seed42
Recipe: FactoryBN_dense_entity_noinfo_start5_min8.json; best epoch 12

Protocol. Same dense_i1 pack and episode split (seed 42). Entity ablation E1: machine-only observations; human/gantry/AGV/buffer dynamics + derived channels masked (indices 13,14,15,17,26). Graph structure kept. Event horizon capped at start ≤ 5 min; Min8 occupancy filter. From-scratch, 50 epochs, empty init_ckpt, unfreeze_last_encoder_blocks=0. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 22: Station-level event reports (start ≤ 5 , entity noinfo). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.802	0.770
	Recall R	0.539	0.553
	F1	0.645	0.644
Stratified recall	Upcoming $R_{up} (s > 0)$	0.056	0.049
	Ongoing $R_{on} (s=0)$	0.895	0.899
Support	n_{true} upcoming	—	389
	n_{true} ongoing	—	566
	n_{true} total	—	955

Table 23: Order remaining time and process-cause heads (start ≤ 5 , entity noinfo).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	21.6	19.4
Process cause (4 active / 6 classes)	Overall accuracy	0.886	0.870
	Macro recall (support>0)	0.906	0.888
	Majority-class baseline (train)		0.409
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.757
	<i>material_shortage</i>	—	0.976
	<i>starved_upstream</i>	—	0.844
	<i>queue_buildup</i>	—	0.976
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.644; report F1 0.644). (2) Best epoch 12/50. Upcoming recall 0.049 on 389 test events.

BNPDFormer Best Checkpoint Metrics

dense_i1_entity_noinfo_start10_min8_cold_ep50_seed42
Recipe: FactoryBN_dense_entity_noinfo_start10_min8.json; best epoch 7

Protocol. Same dense_i1 pack and episode split (seed 42). Entity ablation E1: machine-only observations; human/gantry/AGV/buffer dynamics + derived channels masked (indices 13,14,15,17,26). Graph structure kept. Event horizon capped at start ≤ 10 min; Min8 occupancy filter. From-scratch, 50 epochs, empty init_ckpt, unfreeze_last_encoder_blocks=0. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 24: Station-level event reports (start ≤ 10 , entity noinfo). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.777	0.759
	Recall R	0.417	0.425
	F1	0.543	0.545
Stratified recall	Upcoming $R_{up} (s > 0)$	0.021	0.020
	Ongoing $R_{on} (s=0)$	0.925	0.896
Support	n_{true} upcoming	—	659
	n_{true} ongoing	—	566
	n_{true} total	—	1225

Table 25: Order remaining time and process-cause heads (start ≤ 10 , entity noinfo).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	27.0	24.7
Process cause (4 active / 6 classes)	Overall accuracy	0.616	0.682
	Macro recall (support>0)	0.675	0.707
	Majority-class baseline (train)		0.409
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.934
	<i>material_shortage</i>	—	1.000
	<i>starved_upstream</i>	—	0.667
	<i>queue_buildup</i>	—	0.226
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.545; report F1 0.544). (2) Best epoch 7/50. Upcoming recall 0.020 on 659 test events.

BNPDFormer Best Checkpoint Metrics

dense_i1_entity_noinfo_start15_min8_cold_ep50_seed42

Recipe: FactoryBN_dense_entity_noinfo_start15_min8.json; best epoch 41

Protocol. Same dense_i1 pack and episode split (seed 42). Entity ablation E1: machine-only observations; human/gantry/AGV/buffer dynamics + derived channels masked (indices 13,14,15,17,26). Graph structure kept. Event horizon capped at start ≤ 15 min; Min8 occupancy filter. From-scratch, 50 epochs, empty init_ckpt, unfreeze_last_encoder_blocks=0. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 26: Station-level event reports (start ≤ 15 , entity noinfo). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.455	0.466
	Recall R	0.494	0.518
	F1	0.474	0.490
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.175	0.127
	Ongoing $R_{\text{on}} (s=0)$	0.973	0.922
Support	n_{true} upcoming	—	765
	n_{true} ongoing	—	566
	n_{true} total	—	1331

Table 27: Order remaining time and process-cause heads (start ≤ 15 , entity noinfo).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	15.2	14.0
Process cause (4 active / 6 classes)	Overall accuracy	0.917	0.917
	Macro recall (support>0)	0.926	0.929
	Majority-class baseline (train)		0.409
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.868
	<i>material_shortage</i>	—	0.976
	<i>starved_upstream</i>	—	0.890
	<i>queue_buildup</i>	—	0.984
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.490; report F1 0.441). (2) Best epoch 41/50. Upcoming recall 0.127 on 765 test events.

BNPDFormer Best Checkpoint Metrics

dense_i1_entity_nocross_start5_min8_cold_ep50_seed42

Recipe: FactoryBN_dense_entity_nocross_start5_min8.json; best epoch 29

Protocol. Same dense_i1 pack and episode split (seed 42). Entity ablation E2: all entity features kept; graph_mode=factory_within_type and mask_cross_type_attention=true (no cross-type edges / attention). Event horizon capped at start ≤ 5 min; Min8 occupancy filter. From-scratch, 50 epochs, empty init_ckpt, unfreeze_last_encoder_blocks=0. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 28: Station-level event reports (start ≤ 5 , entity nocross). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.759	0.750
	Recall R	0.696	0.698
	F1	0.726	0.723
Stratified recall	Upcoming $R_{up} (s > 0)$	0.224	0.199
	Ongoing $R_{on} (s=0)$	0.939	0.907
Support	n_{true} upcoming	—	336
	n_{true} ongoing	—	766
	n_{true} total	—	1102

Table 29: Order remaining time and process-cause heads (start ≤ 5 , entity nocross).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	19.1	17.8
Process cause (4 active / 6 classes)	Overall accuracy	0.862	0.862
	Macro recall (support>0)	0.870	0.864
	Majority-class baseline (train)	0.409	
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.956
	<i>material_shortage</i>	—	0.963
	<i>starved_upstream</i>	—	0.882
	<i>queue_buildup</i>	—	0.653
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.723; report F1 0.716). (2) Best epoch 29/50. Upcoming recall 0.199 on 336 test events.

BNPDFormer Best Checkpoint Metrics

dense_i1_entity_nocross_start10_min8_cold_ep50_seed42
Recipe: FactoryBN_dense_entity_nocross_start10_min8.json; best epoch 32

Protocol. Same dense_i1 pack and episode split (seed 42). Entity ablation E2: all entity features kept; graph_mode=factory_within_type and mask_cross_type_attention=true (no cross-type edges / attention). Event horizon capped at start ≤ 10 min; Min8 occupancy filter. From-scratch, 50 epochs, empty init_ckpt, unfreeze_last_encoder_blocks=0. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 30: Station-level event reports (start ≤ 10 , entity nocross). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.625	0.596
	Recall R	0.589	0.593
	F1	0.606	0.595
Stratified recall	Upcoming $R_{up} (s > 0)$	0.233	0.157
	Ongoing $R_{on} (s=0)$	0.922	0.893
Support	n_{true} upcoming	—	604
	n_{true} ongoing	—	768
	n_{true} total	—	1372

Table 31: Order remaining time and process-cause heads (start ≤ 10 , entity nocross).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	15.2	13.9
Process cause (4 active / 6 classes)	Overall accuracy	0.831	0.841
	Macro recall (support>0)	0.846	0.851
	Majority-class baseline (train)		0.409
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.978
	<i>material_shortage</i>	—	1.000
	<i>starved_upstream</i>	—	0.844
	<i>queue_buildup</i>	—	0.581
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.595; report F1 0.571). (2) Best epoch 32/50. Upcoming recall 0.157 on 604 test events.

BNPDFormer Best Checkpoint Metrics

dense_i1_entity_nocross_start15_min8_cold_ep50_seed42
 Recipe: FactoryBN_dense_entity_nocross_start15_min8.json; best epoch 41

Protocol. Same dense_i1 pack and episode split (seed 42). Entity ablation E2: all entity features kept; graph_mode=factory_within_type and mask_cross_type_attention=true (no cross-type edges / attention). Event horizon capped at start ≤ 15 min; Min8 occupancy filter. From-scratch, 50 epochs, empty init_ckpt, unfreeze_last_encoder_blocks=0. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 32: Station-level event reports (start ≤ 15 , entity nocross). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.518	0.518
	Recall R	0.561	0.616
	F1	0.539	0.563
Stratified recall	Upcoming $R_{up} (s > 0)$	0.247	0.205
	Ongoing $R_{on} (s=0)$	0.908	0.892
Support	n_{true} upcoming	—	708
	n_{true} ongoing	—	770
	n_{true} total	—	1478

Table 33: Order remaining time and process-cause heads (start ≤ 15 , entity nocross).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	16.5	15.3
Process cause (4 active / 6 classes)	Overall accuracy	0.798	0.801
	Macro recall (support>0)	0.807	0.811
	Majority-class baseline (train)		0.409
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.971
	<i>material_shortage</i>	—	1.000
	<i>starved_upstream</i>	—	0.814
	<i>queue_buildup</i>	—	0.460
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.563; report F1 0.514). (2) Best epoch 41/50. Upcoming recall 0.205 on 708 test events.

BNPDFormer Best Checkpoint Metrics

dense_i1_entity_machineonly_start5_min8_cold_ep50_seed42

Recipe: FactoryBN_dense_entity_machineonly_start5_min8.json; best epoch 12

Protocol. Same dense_i1 pack and episode split (seed 42). Entity ablation E3: E1+E2 combined (machine observations only + no cross-type relations). Closest to machine-only factory logging. Event horizon capped at start ≤ 5 min; Min8 occupancy filter. From-scratch, 50 epochs, empty init_ckpt, unfreeze_last_encoder_blocks=0. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 34: Station-level event reports (start ≤ 5 , entity machineonly). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.829	0.798
	Recall R	0.554	0.549
	F1	0.664	0.650
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.043	0.033
	Ongoing $R_{\text{on}} (s=0)$	0.930	0.903
Support	n_{true} upcoming	—	389
	n_{true} ongoing	—	566
	n_{true} total	—	955

Table 35: Order remaining time and process-cause heads (start ≤ 5 , entity machineonly).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	25.2	23.3
Process cause (4 active / 6 classes)	Overall accuracy	0.704	0.699
	Macro recall (support>0)	0.680	0.676
	Majority-class baseline (train)	0.409	
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.816
	<i>material_shortage</i>	—	0.988
	<i>starved_upstream</i>	—	0.899
	<i>queue_buildup</i>	—	0.000
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.650; report F1 0.650). (2) Best epoch 12/50. Upcoming recall 0.033 on 389 test events.

BNPDFormer Best Checkpoint Metrics

dense_i1_entity_machineonly_start10_min8_cold_ep50_seed42

Recipe: FactoryBN_dense_entity_machineonly_start10_min8.json; best epoch 11

Protocol. Same dense_i1 pack and episode split (seed 42). Entity ablation E3: E1+E2 combined (machine observations only + no cross-type relations). Closest to machine-only factory logging. Event horizon capped at start ≤ 10 min; Min8 occupancy filter. From-scratch, 50 epochs, empty init_ckpt, unfreeze_last_encoder_blocks=0. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 36: Station-level event reports (start ≤ 10 , entity machineonly). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.818	0.820
	Recall R	0.417	0.424
	F1	0.552	0.559
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.023	0.014
	Ongoing $R_{\text{on}} (s=0)$	0.923	0.901
Support	n_{true} upcoming	—	659
	n_{true} ongoing	—	566
	n_{true} total	—	1225

Table 37: Order remaining time and process-cause heads (start ≤ 10 , entity machineonly).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	28.6	26.8
Process cause (4 active / 6 classes)	Overall accuracy	0.700	0.736
	Macro recall (support>0)	0.710	0.730
	Majority-class baseline (train)	0.409	
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.956
	<i>material_shortage</i>	—	1.000
	<i>starved_upstream</i>	—	0.835
	<i>queue_buildup</i>	—	0.129
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.559; report F1 0.559). (2) Best epoch 11/50. Upcoming recall 0.014 on 659 test events.

BNPDFormer Best Checkpoint Metrics

dense_i1_entity_machineonly_start15_min8_cold_ep50_seed42

Recipe: FactoryBN_dense_entity_machineonly_start15_min8.json; best epoch 14

Protocol. Same dense_i1 pack and episode split (seed 42). Entity ablation E3: E1+E2 combined (machine observations only + no cross-type relations). Closest to machine-only factory logging. Event horizon capped at start ≤ 15 min; Min8 occupancy filter. From-scratch, 50 epochs, empty init_ckpt, unfreeze_last_encoder_blocks=0. Checkpoint metric is will15 F1. Numbers from last_metrics.json.

Table 38: Station-level event reports (start ≤ 15 , entity machineonly). Checkpoint metric is will15 F1.

Group	Metric	Validation	Test
Report / will15 (primary)	Precision P	0.787	0.802
	Recall R	0.387	0.400
	F1	0.519	0.534
Stratified recall	Upcoming $R_{\text{up}} (s > 0)$	0.021	0.021
	Ongoing $R_{\text{on}} (s=0)$	0.936	0.912
Support	n_{true} upcoming	—	765
	n_{true} ongoing	—	566
	n_{true} total	—	1331

Table 39: Order remaining time and process-cause heads (start ≤ 15 , entity machineonly).

Layer	Metric	Validation	Test
Order remaining time	MAE(<i>remain_len</i>) (min)	23.0	21.2
Process cause (4 active / 6 classes)	Overall accuracy	0.724	0.744
	Macro recall (support>0)	0.728	0.735
	Majority-class baseline (train)	0.409	
	Valid samples n_{cause}	—	579
Per-class recall	<i>transport_delay</i>	—	0.875
	<i>material_shortage</i>	—	1.000
	<i>starved_upstream</i>	—	0.865
	<i>queue_buildup</i>	—	0.202
	<i>blocked_downstream</i>	no support on test	
	<i>high_utilization</i>	no support on test	

Notes. (1) Val primary numbers are report_* at the best will15 epoch; test primary is will15_* (F1 0.534; report F1 0.534). (2) Best epoch 14/50. Upcoming recall 0.021 on 765 test events.

Entity ablation vs full baseline

Test will15, same dense_i1 split, seed 42

Reading. Full rows are the retained 12_3 runs (up to 100 epochs). Entity arms are cold-start, 50 epochs. The planned Full-Scratch-50 (E0) control was *not* run, so Δ vs full mixes structure removal and training-budget differences. `noinfo/machineonly` change the observed event pool (n_{true} differs from full/`nocross`); compare those two against each other first, and treat full deltas as directional only. R_{up} is report upcoming recall.

Table 40: Test will15 and auxiliary heads. Bold F1 is the checkpoint metric.

Arm	Start	P	R	F1	R_{up}	R_{on}	remain MAE
full	≤ 5	0.794	0.817	0.805	0.539	0.937	10.9
noinfo	≤ 5	0.770	0.553	0.644	0.049	0.899	19.4
nocross	≤ 5	0.750	0.698	0.723	0.199	0.907	17.8
machineonly	≤ 5	0.798	0.549	0.650	0.033	0.903	23.3
full	≤ 10	0.806	0.761	0.783	0.528	0.936	11.2
noinfo	≤ 10	0.759	0.425	0.545	0.020	0.896	24.7
nocross	≤ 10	0.596	0.593	0.595	0.157	0.893	13.9
machineonly	≤ 10	0.820	0.424	0.559	0.014	0.901	26.8
full	≤ 15	0.806	0.742	0.772	0.534	0.925	9.7
noinfo	≤ 15	0.466	0.518	0.490	0.127	0.922	14.0
nocross	≤ 15	0.518	0.616	0.563	0.205	0.892	15.3
machineonly	≤ 15	0.802	0.400	0.534	0.021	0.912	21.2

Notes. (1) E1 `noinfo` and E3 `machineonly` share the masked-observation event pool ($n_{\text{who}}=955/1225/1331$); E2 `nocross` matches full (1102/1372/1478). (2) Across starts, removing entity dynamics (E1) hurts early warning most: R_{up} collapses to 0.049/0.020/0.127 vs full ≈ 0.53 . (3) Removing only cross-type relations (E2) keeps more upcoming recall (0.199/0.157/0.205) than E1/E3, but still drops F1 by ~ 0.08 – 0.21 vs full and damages *queue_buildup* cause recall (0.653/0.581/0.460). (4) E3 is not much worse than E1 on F1 at $\text{Start} \leq 5/10$ (0.650 vs 0.644; 0.559 vs 0.545), so once entity dynamics are gone, further cutting cross-type edges adds little; the extra damage shows mainly in remain MAE and *queue_buildup*.